BOWEN JIANG

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EDUCATION

Duke University — **Durham, NC**
B.S. in Biomedical Engineering and Electrical & Computer Engineering — *Expected May 2027*

- **GPA:** 3.9/4.0 | **Coursework:** Deep Neural Nets, Connectomic Neuromodulation, Signals & Systems, Microelectronic Devices, Data Structures, Quantitative Physiology, Linear Algebra, Computer Architecture, Biochip Engineering, Imaging Systems, OS

SKILLS

Languages: Python, C++, Rust, Java, Swift, R, MATLAB
Framework & Tools: React, Node.js, PyTorch, OpenCV, scikit-learn, Pandas, REST APIs, HTML/CSS
Cloud & Developer Tools: AWS, Nebius Cloud, Docker, Git, Linux, SLURM/HPC

PROJECTS

Wind — **Durham, NC**
Co-Founder — *Aug 2026 – Present*

- Built a usage-based wallet integrating Stripe and multiple AI model-routing APIs to shift inference costs from devs to individual users
- Designed OAuth authentication, Redis-backed metering, and per-user usage infrastructure for secure billing and multi-model access

TryIt — **San Francisco, CA**
Co-Founder — *Jul 2025 – Present*

- Built a full-stack, Gemini-based, and personalized fashion mobile and web app enabling prompt-based and in-store virtual try-on
- Launched a Duke University Store pilot, engineering SKU mapping and privacy-preserving image workflows for in-store deployment

CalendarOCR — **San Francisco, CA**
Co-Founder — *Sep 2025 – Present*

- Built and shipped an Expo-based iOS app combining OCR, LLM extraction, and calendar APIs to convert event posters into structured calendar events; reached 120+ downloads, 10.2% signup conversion, and 5% paid conversion

INDUSTRY EXPERIENCE

Amazon — **Sunnyvale, CA**
SDE Intern, Devices — *Sep 2026 – Nov 2026*

- Led end-to-end development of a Fire TV UI feature, migrating functionality from Android to a proprietary OS using C++, collaborating across teams, and owning technical design through final integration

Science Corporation — **Alameda, CA**
Protein Engineering Intern — *Jun 2026 – Aug 2026*

- Built automation workflows on the Galago and PLR platforms to screen ~20,000 opsin variants across high-throughput experiments
- Automated colony picking to 25 384-well plates and developed a Nanopore pipeline for large-scale library sequencing

Tetsuwan Scientific — **San Francisco, CA**
Lab Automation Intern — *May 2025 – Aug 2025*

- Built Python automation pipelines for DNA sequencing and bacterial plating on Opentrons, reducing technician hands-on time by 90%
- Developed an OpenCV system for automated colony detection and picking with 95% recognition accuracy and 3× faster plate processing
- Launched LabAutoWiki.org, the go-to community site for lab automation information, scaling to 12+ contributors and 50+ pages

RESEARCH EXPERIENCE

Navid Lab — **Durham, NC**
Student Researcher — *Aug 2023 – May 2025*

- Built a PyTorch training pipeline using ESM2 and LoRA to predict protein variant fitness across ~700,000 variants
- Developed primal-dual constraint learning with 5-fold validation, improving Spearman correlation by ~2% over baseline models

Chory Lab — **Durham, NC**
Student Researcher — *Aug 2024 – Present*

- Built PLR-based Hamilton and Opentrons protein evolution workflows with 72-hour autonomous operation and real-time monitoring
- Contributed PLR visualization modules and optimized PCR/Qubit automation, reducing reaction costs while increasing throughput

PUBLICATIONS

Boileau R.M., Golas S.M., Ma Q., **Jiang B.**, *et al* (2026). "An autonomous system for multi-objective continuous evolution at scale." ***bioRxiv (preprint)***. https://doi.org/10.64898/2026.03.02.709196

Barreto A., **Jiang B.**, Burt M., Dimitrakakis N., Musah S (2026). "Discovery of kidney disease targets using multimodal human podocyte injury models." ***Stem Cell Reports.*** https://doi.org/10.1016/j.stemcr.2026.102868